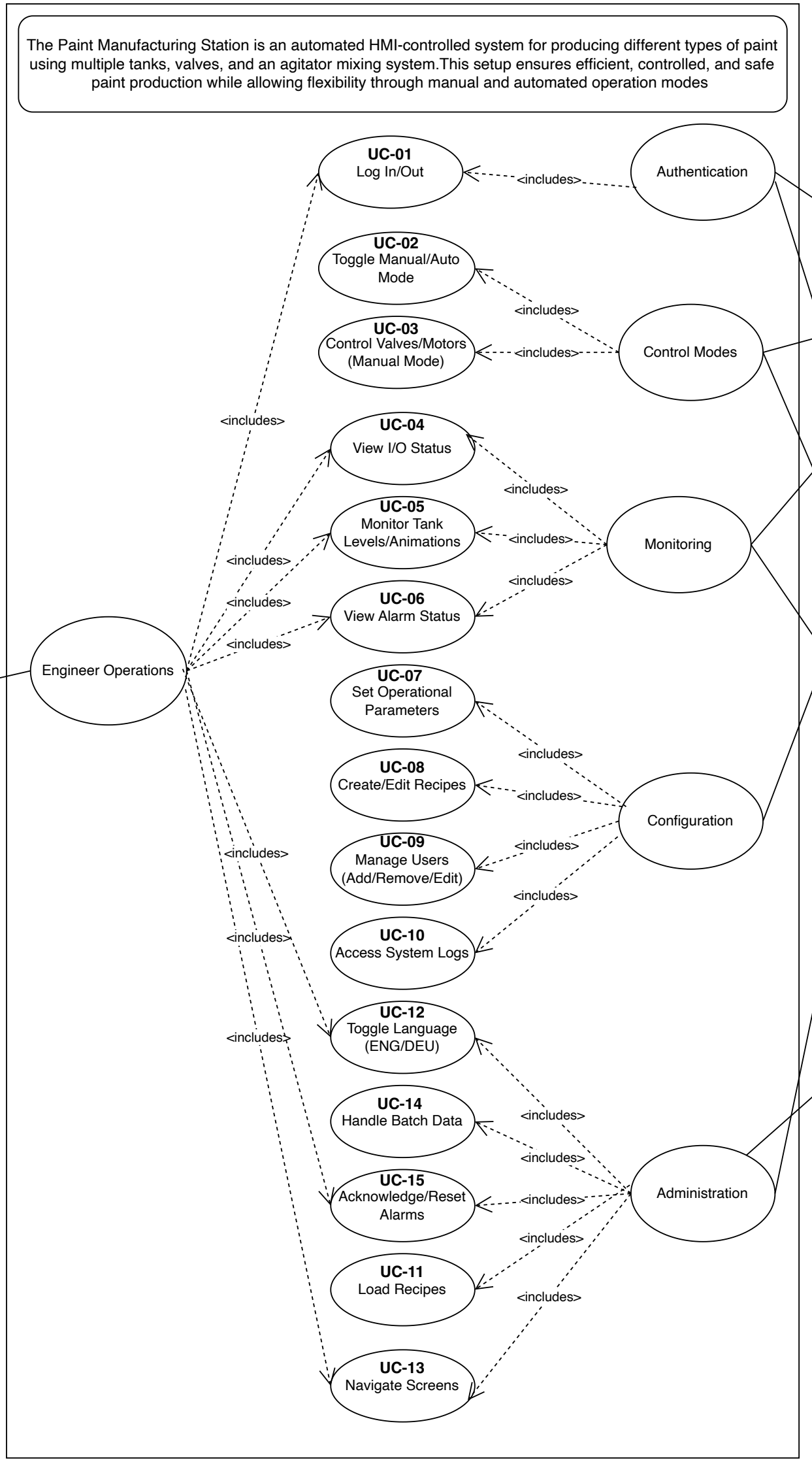
Use Case Diagram - Paint Manufacturing Station

Note: HMI Project done using SIEMENS WinCC Advanced Software v17



Engineer

An Engineer is responsible for system monitoring, diagnostics, and user management. They can view system statuses, logs, alarms, and batch data but cannot directly control

Administrator

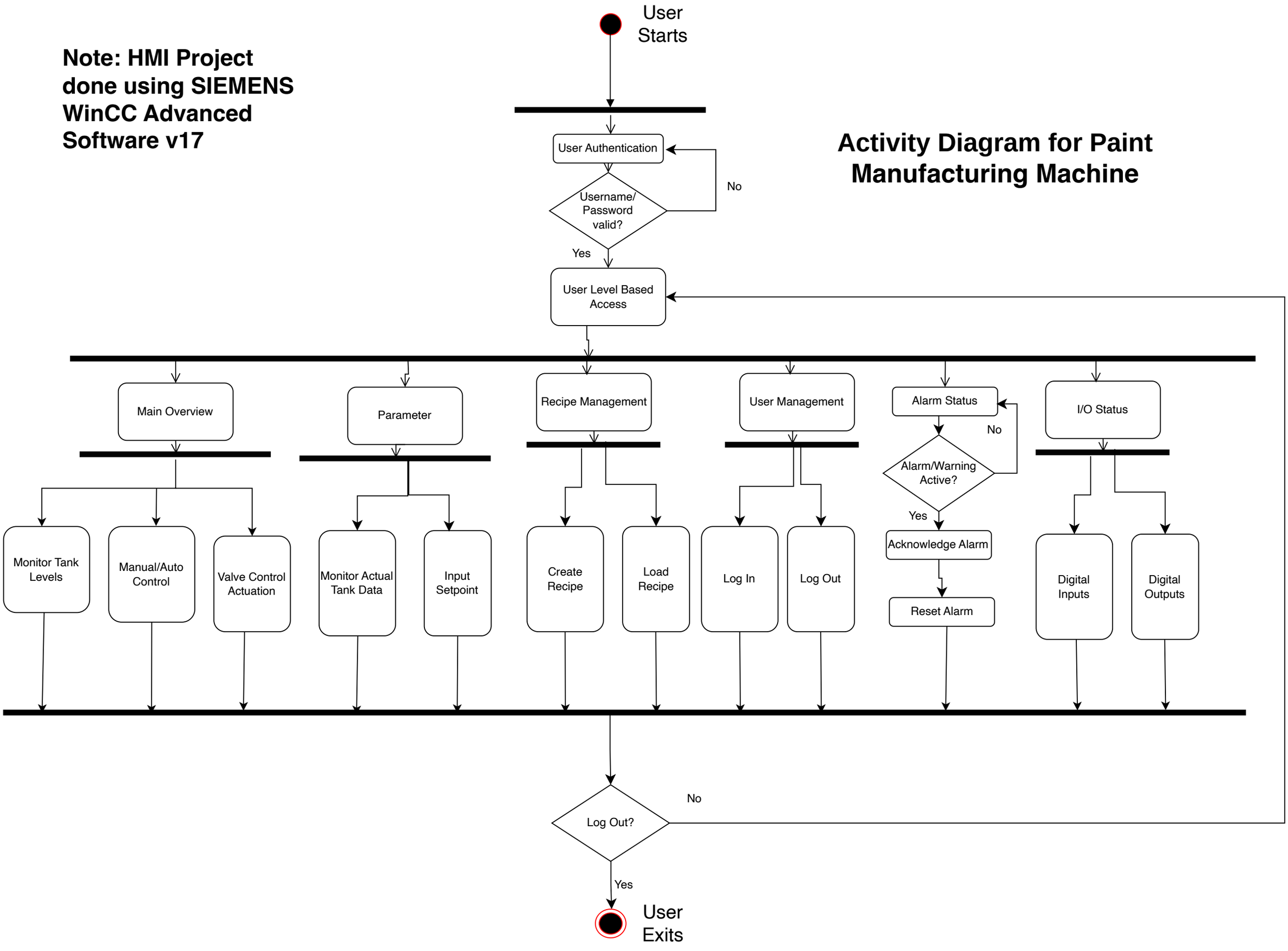
An Administrator has full system access and privileges. They manage users, configure system settings, control all operational

Operator

An Operator interacts with the system to control and monitor processes in real time. They can switch between manual and auto modes, adjust operational parameters, control valves and motors, and acknowledge alarms, ensuring smooth plant operations.

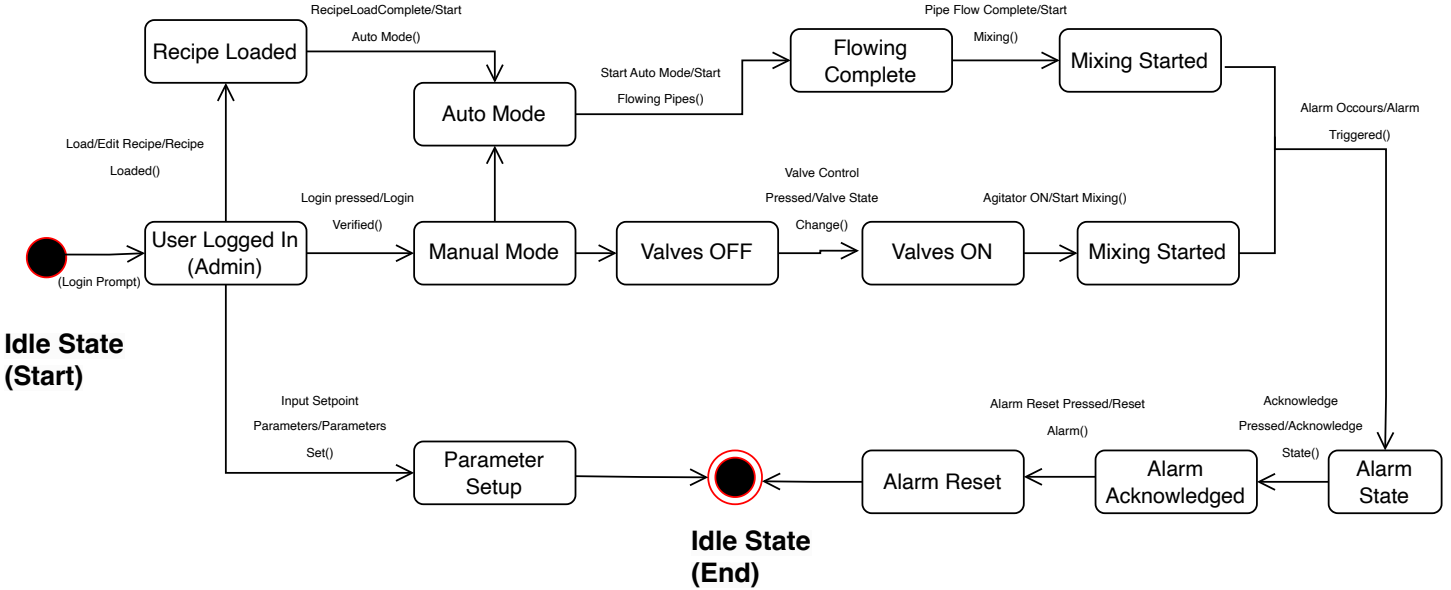
Note: HMI Project done using SIEMENS WinCC Advanced Software v17

Activity Diagram for Paint Manufacturing Machine



Note: HMI Project done using SIEMENS WinCC Advanced Software v17

State Diagram - Paint Manufacturing Station



Design Considerations of the Mock-up

To ensure an intuitive and efficient **Human-Machine Interface (HMI)**, various **design considerations** have been incorporated into the interface, linking them to specific **use cases** and activity flows. The following table outlines how these design principles enhance usability for each use case.

Design Consideration	UI Option	Mapped Use Cases	Description
Low Cognitive Load	Minimalistic dashboard with only essential controls and status indicators	UC-02 (Toggle Manual/Auto Mode), UC-04 (View I/O Status), UC-05 (Monitor Tank Levels)	The interface should avoid clutter, ensuring only relevant data (e.g., tank levels, alarms, I/O status) is displayed at a time. Manual/Auto mode toggle should be prominent and easy to switch.
Pre-Attentive Processing with Pop-ups	Flashing alarm icon with high-contrast pop-ups for warnings/errors	UC-06 (View Alarm Status), UC-15 (Acknowledge/Reset Alarms)	Critical alarms should trigger large, red pop-ups with concise messages and reset options to grab user attention instantly.
Distinguishable Colors for Similar Objects	Different colors for manual/auto mode and motor/valve states	UC-02 (Toggle Manual/Auto Mode), UC-03 (Control Valves/Motors)	Manual mode (yellow) vs. Auto mode (green); active motors (blue), inactive (gray). Helps quick recognition of system status.
Segmented Control	Tabs and buttons for different sections (Overview, Recipes, Parameters)	UC-13 (Navigate Screens), UC-07 (Set Operational Parameters)	The navigation bar should use clear tabs to group related actions, making it easy for users to switch between screens without confusion.
Easy Recognition of Widgets and Buttons	Clear icons for start/stop/reset actions with consistent positioning	UC-03 (Control Valves/Motors), UC-05 (Monitor Tank Levels)	Every button should have a recognizable icon (e.g., Rotating Motor Pumps for each Pigment Pipe) to eliminate ambiguity.
Affordance (Easy Recognition of Actions)	Toggle switches, dropdowns, and buttons designed to indicate interactivity	UC-02 (Toggle Manual/Auto Mode), UC-12 (Toggle Language)	Auto/manual mode should have a switch that visibly changes position. Language toggles should use flags/icons rather than just text.
Direct Manipulation	Interactive elements like tank level indicators and motor status buttons	UC-05 (Monitor Tank Levels), UC-03 (Control Valves/Motors)	Users should be able to click directly on the tank graphic to see more details. Motors should be controlled by pressing on their graphical representation.

The System Should Have Limited States	Clearly defined states (Stopped, Running, Maintenance, Alarm) with a dedicated display	UC-02 (Toggle Manual/Auto Mode), UC-06 (View Alarm Status)	The interface should display only valid system states, preventing unintended transitions. A "Manual" state should only be triggered by pressing "Manual" in COntrol Panel.
Icon Grouping with Text Combinations	Icons with labels for better recognition (e.g., "Save Recipe" button with a disk icon)	UC-08 (Create/Edit Recipes), UC-11 (Save/Load Recipes)	Buttons like "Save Recipe" should combine icons and text to ensure users recognize their function immediately.

List of Use Cases in the Project

Use Case (UC)	UI Element	Design Considerations
UC-01: Log In/Out	Login Screen	Colour Combinations, Recognition of Response After Input, Simple Interaction Options
UC-02: Toggle Manual/Auto Mode	Mode Toggle Button	Easy Recognition of Widgets, Limited States, Affordance
UC-03: Control Valves/Motors (Manual Mode)	Valve/Motor Control Panel	Direct Manipulation, Time Required for Response, Recognition of Response After Input
UC-04: View I/O Status (Digital Inputs/Outputs)	I/O Status Display	Colour Combinations, Low Cognitive Load, Coordination of Content
UC-05: Monitor Tank Levels/Animations	Tank Level Graphical Display	Perception & Evaluation, Shape of Sub-Objects, Similarity & Continuity
UC-06: View Alarm Status	Alarm Popup & Acknowledgment	Pre-Attentive Processing, Reflexive Processes, Recognition of Response After Input
UC-07: Set Operational Parameters (Flow Rate, Viscosity, etc.)	Parameter Input Fields	Simple Interaction Options, Shape of Sub-Objects, Recognition of Response After Input
UC-08: Create/Edit Recipes (e.g., "Emerald Green")	Recipe Management Screen	Segmented Control, Coordination of Content, Affordance
UC-09: Manage Users (Add/Remove/Edit)	User Management Panel	Icon Grouping, Recognition of Response After Input, Simple Interaction Options
UC-11: Save/Load Recipes	Save/Load Recipe Interface	Lower Complexity, Recognition of Response After Input, Simple Interaction Options
UC-12: Toggle Language (ENG/DEU)	Language Toggle Button	Easy Recognition of Widgets, Affordance, Perception & Evaluation
UC-13: Navigate Screens (Main Overview, Parameter, Recipe, etc.)	Navigation Menu	Direct Manipulation, Segmented Control, Lower Complexity
UC-14: Handle Batch Data (Recipe ID, Batch Size, Timestamp)	Batch Data Entry Form	Shape of Sub-Objects, Simple Interaction Options, Recognition of Response After Input
UC-15: Acknowledge/Reset Alarms	Alarm Reset Button	Reflexive Processes, Recognition of Response After Input, Time Required for Response

Design Approach

1. Alarm Pop-Up (UC-06: View Alarm Status, UC-15: Acknowledge/Reset Alarms)

- **Affordance:** The alarm pop-up clearly indicates an urgent issue requiring user action.
- **Design Considerations:**
    1. **Pre-Attentive Processing:** A red exclamation icon ensures quick recognition.
    2. **Recognition of Response After Input:** Confirmation message appears after acknowledging/resetting the alarm.
    3. **Reflexive Processes:** A reset button directly removes the alarm message.

2. Toggle Manual/Auto Mode (UC-02: Toggle Manual/Auto Mode)

- **Affordance:** The interface clearly indicates that users can switch between two modes.
- **Design Considerations:**
    1. **Limited States:** The system only allows "Manual" or "Auto" mode, preventing unnecessary confusion.
    2. **Segmented Control:** The toggle button is placed in a logical section for easy access.
    3. **Easy Recognition:** The button shape and color make it distinguishable.

3. Tank Level Display (UC-05: Monitor Tank Levels/Animations)

- **Affordance:** The visual representation allows users to monitor levels without reading text values.
- **Design Considerations:**
    1. **Similarity and Continuity:** The tank animation smoothly transitions based on real-time data.
    2. **Colour Combinations for Text and Background:** Uses contrasting colors to differentiate safe/critical levels.
    3. **Perception & Evaluation:** Users can immediately understand if the level is too high or too low.

4. Recipe Selection Menu (UC-08: Create/Edit Recipes, UC-11: Save/Load Recipes)

- **Affordance:** A drop-down selection allows users to pick a recipe instead of manually entering data.
- **Design Considerations:**
    1. **Simple Interaction Options:** Allows easy selection instead of typing.
    2. **Coordination of Content in Window Screen:** The menu is positioned logically near recipe settings.
    3. **Lower Complexity:** Only relevant options are shown, reducing unnecessary clutter.

5. Navigation Menu (UC-13: Navigate Screens)

- **Affordance:** Users can switch between different screens easily.
- **Design Considerations:**
    1. **Icon Grouping with Text Combinations:** Icons help users quickly recognize sections.
    2. **Direct Manipulation:** Clicking a tab immediately changes the screen.
    3. **Shape of Sub-Objects:** The navigation menu is designed with clearly structured drop-down lists.

6. Language Toggle (UC-12: Toggle Language ENG/DEU)

- **Affordance:** A simple button allows users to switch between English and German.
- **Design Considerations:**
    1. **Easy Recognition of Widgets and Buttons:** Uses country flags/icons instead of just text.
    2. **Affordance:** Users instantly know this button changes the interface language.
    3. **Perception & Evaluation:** When the language is changed, all interface text updates immediately.

7. Valve/Motor Control Panel (UC-03: Control Valves/Motors in Manual Mode)

- **Affordance:** Buttons and indicators make it clear how to operate valves/motors.
- **Design Considerations:**
    1. **Direct Manipulation:** Users can click buttons to start/stop motors or open/close valves.
    2. **Time Required for Response:** Status updates are instant, providing immediate feedback.
    3. **Recognition of Response After Input:** A **green "ON" indicator** confirms activation, while **red "OFF"** confirms deactivation.

8. User Authentication Screen (UC-01: Log In/Out)

- **Affordance:** The interface clearly indicates where users should enter their credentials.
- **Design Considerations:**
    1. **Colour Combinations for Text and Background:** Text fields have a high contrast with the background to improve readability.
    2. **Recognition of Response After Input:** Displays error messages like *"Incorrect Password"* if login fails.
    3. **Simple Interaction Options:** Provides a **password field with hidden input** and a **"Show Password" toggle** for usability.

Test of the System

This section provides test cases to verify the system's functionality, covering user authentication, control operations, and responsiveness. Each test includes steps and expected outcomes for easy validation.

Test Case ID	Test Description	Test Steps	Expected Result
TC-01	User Login	1. Open login screen. 2. Enter username & password. 3. Click "Login".	System grants access if credentials are correct, else shows error.
TC-02	Toggle Manual/Auto Mode	1. Navigate to control mode. 2. Click the toggle button.	System switches mode and updates display.
TC-03	Control Valves/Motors in Manual Mode	1. Select a valve/motor. 2. Click "Start/Stop".	The selected component responds accordingly.
TC-04	View I/O Status	1. Open I/O status screen.	Display updates real-time with correct values.
TC-05	Monitor Tank Levels	1. Navigate to the monitoring screen. 2. Observe tank level animations.	Display shows correct tank levels and animations update.
TC-06	View Alarm Status	1. Trigger an alarm condition. 2. Open the alarm screen.	Alarm pop-up appears with correct status.
TC-07	Set Operational Parameters	1. Open parameter settings. 2. Enter new values (e.g., flow rate). 3. Save changes.	System updates and applies new settings.
TC-08	Create/Edit Recipe	1. Open recipe management. 2. Create/edit a recipe. 3. Save changes.	Recipe is saved and listed correctly.
TC-09	Acknowledge & Reset Alarms	1. Open alarm screen. 2. Click "Acknowledge" and then "Reset".	Alarm disappears after reset.
TC-11	Save & Load Recipe	1. Open recipe management. 2. Save a new recipe. 3. Load the saved recipe.	The saved recipe loads with correct parameters.
TC-12	Toggle Language (ENG/DEU)	1. Go to settings. 2. Select a different language (e.g., from English to German).	All UI elements update to the selected language.
TC-13	Navigate Between Screens	1. Use navigation buttons to switch between screens (e.g., Main, Recipe, Parameter).	The UI updates correctly without lag or errors.
TC-14	Handle Batch Data	1. Enter batch details (ID, size, timestamp). 2. Save and check batch history.	The batch data is saved and correctly displayed in history.
TC-17	Alarm Notification Visibility	1. Trigger an alarm. 2. Check if alarm pop-up appears with icon.	Alarm notification is clearly visible with appropriate alerts.
TC-18	System Responsiveness	1. Perform multiple actions quickly (e.g., navigate, enter values).	No UI freezing or unresponsive behavior occurs.
TC-19	Incorrect Input Handling	1. Enter invalid data (e.g., letters in numerical fields).	The system rejects input and displays an error message.